Myungjin Lee

Post-Master Researcher · PRISM Lab, Dongguk University

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Research Overview

I develop interpretable graph models for molecular property prediction in drug discovery. I use bulk and single-cell transcriptomics to study cellular senescence and aging-related metabolic dysfunction-associated steatotic liver disease (MASLD).

Graph Representation Learning · Molecular Property Prediction · Transcriptomic Analysis · Explainable AI

Education

Dongguk University · Seoul, Republic of Korea

M.E. in Computer Science and Artificial Intelligence — Mar 2024 – Aug 2026

Major: AI in Healthcare and Medicine · GPA: 4.04/4.5 · Advisor: Dr. Sangsoo Lim

Thesis: *Graph Substructure-Based Junction Tree Modeling for Drug ADMET Prediction*

Induk University · Seoul, Republic of Korea

B.E. in Industrial & Management Engineering — Mar 2023 – Feb 2024

Advanced Major Program · Convergence Minor in Deep Learning · GPA: 4.43/4.5; Graduated with honors

A.E. in Industrial & Management Engineering — Mar 2020 – Feb 2023

GPA: 4.41/4.5 (major: 4.42/4.5) · Valedictorian

Manuscripts

1. **Lee M.**, Mo J., Kang M., and Lim S. *(Under review).* SJoINT: Substructure-Driven Junction Tree for Interpretable Molecular Property Prediction.
2. Mo J., **Lee M.**, Lee S., and Lim S. *(Under review).* Cross-view structural consensus for interpretable molecular property prediction.
3. Lee G., Byeon S.K., Olaleye O.E., Han Y., **Lee M.**, Vizioli M.G., Ogrodnik M., Pirius N., Sales Gomez L., Victorelli S., Khosla S., Pandey A., Passos J., Lim S., and Jurk D. *(In preparation).* Lipid abundance and lysophospholipid signaling regulate the senescence-associated secretory phenotype.

Publications

1. **Lee M.**, Mo J., and Lim S. (2026). Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning. *Proceedings of the Korea Computer Congress (KCC)*, KIISE, pp. 291–293. [Paper]

Research Funding

Research Subsidies for Master's Students

Principal investigator — Sep 2024 – Aug 2025

IITP (MSIT) · Grant No. S-2025-A0434-00016 · Advisor: Dr. Sangsoo Lim

Multi-task molecular graph representation learning with functional-group-based junction trees.

Research Experience

PRISM Lab, Dongguk University

Post-Master Researcher Sep 2026 – Present

Graduate Researcher Jan 2024 – Aug 2026

Advisor: Dr. Sangsoo Lim · Seoul, Republic of Korea

- Developed SJoINT for interpretable molecular property prediction (first author); contributed experiments, figures, and manuscript preparation to MAGNET (second author).
- Investigated lipid metabolism and cellular senescence through parallel transcriptomic analyses in the aging-related MASLD project and a related lipid/SASP study with Mayo Clinic collaborators.

Research Projects

Open Platform Center for AI-Powered Drug Discovery and Analytics

Participating researcher Jul 2025 – Present

PI: Dr. Minho Lee, Dongguk University · Project RS-2025-18732993

- Benchmarked 5 repurposing models on curated PrimeKG data (3,379 drugs; 935 diseases; 190 held-out test diseases); developed DC-VGAE and a performance-weighted ensemble.
- Benchmarked 13 molecular property models on 9 MoleculeNet datasets using common molecule-level splits across 5 seeds.
- Containerized model implementations for reproducible deployment; 19 models and the curated PrimeKG graph were registered on KBDI.

Multimodal AI for Target Discovery and Drug Validation in Aging-Related MASLD

Participating researcher Mar 2025 – Feb 2026

PI: Dr. Sangsoo Lim, Dongguk University · Project RS-2025-00560523

Collaborator: Dr. Gung Lee, Mayo Clinic, USA

- Curated public bulk and single-cell liver transcriptomes into a unified schema for age, diet, disease-stage, and fibrosis metadata.
- Curated lipid-related pathway terms and analyzed age-associated expression, pathway enrichment, and cell-type-specific senescence signatures.

Poster Presentations

BIOINFO/GIW ISCB-Asia 2026 Nov 2026

Cross-attentive Integration of Atom-level and Junction-tree Representations for Molecular Property Prediction

Scheduled · Yonsei University, Seoul, Republic of Korea

Korea Computer Congress (KCC) 2026 Jun 2026

Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning

ICC Jeju, Republic of Korea

BIOINFO 2025 Oct 2025

Weighted Junction-Tree Nodes for Enhanced Interpretability in ADMET Tasks

Suwon Convention Center, Republic of Korea

Patents and Software

System and Method for Predicting Olfactory Characteristics of Odor Mixture Data
Korean patent application 10-2025-0003146 · *Pending* 2025

An Embedding Method via Multi-view Graph Integration of Molecular Structures
Software registration C-2025-040207 · Korea Copyright Commission 2025

Deep Learning Model for Similarity Prediction of Graph-based Mixtures
Software registration C-2024-042597 · Korea Copyright Commission 2024

Honors and Awards

Highest Academic Achievement Award 2023
Induk University, Republic of Korea

Merit-based Academic Scholarship 2020–2023
Induk University, Republic of Korea; awarded every semester, including the GRAPE Talent Scholarship.

Research Competitions

Samsung AI Challenge 2024 Aug 2024 – Sep 2024
Team DAI · Samsung Electronics SAIT / DACON
11th place (top 10%); did not advance to the finals.

- Developed machine learning force fields with uncertainty quantification to predict atomic energies and forces for semiconductor material simulation.

DREAM Olfactory Mixtures Prediction Challenge Apr 2024 – Aug 2024
International DREAM Challenge · Sage Bionetworks

- Developed a graph neural network to predict olfactory similarity between molecular mixtures.

Technical Skills

Programming	Python, R (basic), Java, C#, JavaScript, SQL
Machine learning	PyTorch, PyTorch Geometric, scikit-learn, Hugging Face Transformers, GNNExplainer
Cheminformatics	RDKit, DeepChem, TorchDrug, Chemprop
Bioinformatics	Scanpy, Seurat; bulk and single-cell RNA-seq; differential expression (DESeq2, PyDESeq2, MAST); GSEA, GSVA, GO enrichment; ssGSEA, AddModuleScore, AUCell
Scientific computing	pandas, NumPy, anndata, NetworkX, seaborn, matplotlib
Tools and databases	Git, Linux, Docker, MySQL

Languages

Korean	Native
English	TOEIC 860 (LC 440 / RC 420); OPIc IH